

Calculus Notes

Companion Study Notes

A compact, concept-first companion to the MA1110 Calculus-I material

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1 Basics

Calculus is the mathematics of change. The foundations used throughout this course are properties of the real numbers, bounds, supremum and infimum, the Archimedean property, density, and absolute value. These ideas provide the language needed for limits and convergence.

Definition. Let $A \subset \mathbb{R}$ be nonempty. A real number α is an *upper bound* of A when

$$x \leq \alpha \quad \text{for every } x \in A.$$

The set is *bounded above* when it has at least one upper bound. It is *not bounded above* when for every $\alpha \in \mathbb{R}$ there exists $x \in A$ with $x > \alpha$.

Definition. A real number β is a *lower bound* of A when $\beta \leq x$ for every $x \in A$. A set is *bounded below* when it has a lower bound; it is bounded when it is both bounded above and bounded below.

Definition. If A is nonempty and bounded above, $\sup A$ is its least upper bound: it is an upper bound and it is no larger than any other upper bound. If A is nonempty and bounded below, $\inf A$ is its greatest lower bound: it is a lower bound and it is no smaller than any other lower bound.

Theorem. Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$. Every nonempty subset of $\mathbb{R}$ that is bounded below has an infimum in $\mathbb{R}$.

1.1 Properties of the Supremum and the Infimum

If $M = \sup S$, then every $x \in S$ satisfies $x \leq M$, and for every $\varepsilon > 0$ there exists $y \in S$ such that

$$M - \varepsilon < y \leq M.$$

Similarly, if $m = \inf S$, then $m \leq x$ for all $x \in S$ and, for every $\varepsilon > 0$, some $y \in S$ satisfies

$$m \leq y < m + \varepsilon.$$

The second property says that elements of the set can be found arbitrarily close to the extremal bound.

Theorem. Archimedean property. If $x, y > 0$, there is $n \in \mathbb{N}$ with $ny > x$. In particular, natural numbers are not bounded above, and for every $x > 0$ there is $n \in \mathbb{N}$ with $0 < 1/n < x$.

Theorem. Density. If $x < y$ are real numbers, there is a rational $r \in \mathbb{Q}$ and an irrational $s \in \mathbb{R} \setminus \mathbb{Q}$ such that

$$x < r < y, \quad x < s < y.$$

Definition. For $x \in \mathbb{R}$, the absolute value is

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

It satisfies $|a| \geq 0$, $|ab| = |a||b|$, $|a/b| = |a|/|b|$ for $b \neq 0$, and the triangle inequality

$$|a + b| \leq |a| + |b|.$$

Also $|a - b| < \varepsilon$ exactly when $a \in (b - \varepsilon, b + \varepsilon)$.

Key result. Useful identities are

$$||a| - |b|| \leq |a - b|, \quad \max\{a, b\} = \frac{a + b + |a - b|}{2}, \quad \min\{a, b\} = \frac{a + b - |a - b|}{2}.$$

These identities later connect absolute value with convergence and with maxima/minima.

Representative example. For $A = (0, 1)$, $\sup A = 1$ and $\inf A = 0$, even though neither endpoint belongs to A . This illustrates why a supremum or infimum is a bound, not necessarily a member of the set.

2 Sequences

A sequence of real numbers is a function $f : \mathbb{N} \rightarrow \mathbb{R}$, written (a_n) . The course treats sequences as the first rigorous setting in which approximation, convergence, divergence, monotonicity, subsequences, and Cauchy behavior are studied.

Definition. A sequence (a_n) *converges* to $a \in \mathbb{R}$ when for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon \quad (n \geq N).$$

We write $a_n \rightarrow a$ or $\lim_{n \rightarrow \infty} a_n = a$.

The quantifier order matters: one first fixes an arbitrary tolerance ε , then chooses an index N after which every term lies inside that tolerance.

Theorem. Uniqueness of the limit. A convergent sequence has only one real limit.

Definition. A sequence is *bounded* if there is $C > 0$ with $|a_n| \leq C$ for every n . It is *unbounded* when every $C > 0$ is exceeded by at least one term.

Theorem. Every convergent real sequence is bounded. The converse is false: the harmonic partial sums are increasing and unbounded even though their terms have the form $a_n = 1 + 1/2 + \cdots + 1/n$.

Theorem. Algebra of convergent sequences. If $a_n \rightarrow a$, $b_n \rightarrow b$, and $\alpha \in \mathbb{R}$, then

$$a_n + b_n \rightarrow a + b, \quad \alpha a_n \rightarrow \alpha a, \quad a_n b_n \rightarrow ab.$$

If $a > 0$ and $a_n > 0$ for all n , then $1/a_n \rightarrow 1/a$.

2.1 Convergence and divergence

Common benchmark sequences are

$$1/n \rightarrow 0, \quad n/(n+1) \rightarrow 1, \quad (-1)^n \text{ diverges}, \quad n \rightarrow +\infty.$$

The sequence $((1)^n)$ is bounded but divergent, showing that divergence need not mean growth without bound.

Representative example. To prove $1/n \rightarrow 0$, given $\varepsilon > 0$ use the Archimedean property to choose $N > 1/\varepsilon$. Then $n \geq N$ implies $1/n \leq 1/N < \varepsilon$.

2.2 Properties of convergent sequences

If $a_n \rightarrow a \neq 0$, then $|a_n|$ is eventually bounded away from zero; for example, there is N with $|a_n| > |a|/2$ for $n \geq N$. If $a_n \geq 0$ for every n and $a_n \rightarrow a$, then $a \geq 0$.

Theorem. Sandwich theorem. If $a_n \rightarrow \alpha$, $b_n \rightarrow \alpha$, and

$$a_n \leq c_n \leq b_n \quad (n \in \mathbb{N}),$$

then $c_n \rightarrow \alpha$.

Representative example. Since $-1 \leq \sin n \leq 1$, division by $n > 0$ gives

$$-1/n \leq \sin n/n \leq 1/n,$$

so $\sin n/n \rightarrow 0$ by the Sandwich theorem.

2.3 Monotone sequences

Definition. A sequence is increasing when $a_n \leq a_{n+1}$, strictly increasing when $a_n < a_{n+1}$, decreasing when $a_n \geq a_{n+1}$, and strictly decreasing when $a_n > a_{n+1}$. A monotone sequence is increasing or decreasing.

Theorem. Monotone Convergence Theorem. An increasing sequence that is bounded above converges. A decreasing sequence that is bounded below converges.

The limit is the supremum of the range of an increasing bounded sequence and the infimum of the range of a decreasing bounded sequence.

Important applications in the parent material include $r^n \rightarrow 0$ for $0 \leq r < 1$, the geometric partial sums

$$1 + r + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r} \rightarrow \frac{1}{1 - r},$$

and recursive sequences such as $a_{n+1} = \frac{1}{2}(a_n + 2/a_n)$, whose monotonicity and lower bound lead to the limit $\sqrt{2}$.

The sequence $(1 + 1/n)^n$ is increasing and bounded above; its limit is denoted by e . Likewise, the partial sums $\sum_{k=0}^n 1/k!$ increase to e .

2.4 Sandwich theorem

The Sandwich theorem is especially effective when direct algebra is awkward but simple upper and lower estimates are available. Besides $\sin n/n \rightarrow 0$, the course uses it for square-root sums and for expressions such as $(a^n + b^n)^{1/n}$.

For $a > 0$, $a^{1/n} \rightarrow 1$. If $0 < a < b$, then

$$(a^n + b^n)^{1/n} \rightarrow b,$$

because $b \leq (a^n + b^n)^{1/n} \leq 2^{1/n}b$ and $2^{1/n} \rightarrow 1$.

2.5 Sequences diverging to $\pm\infty$

Definition. We write $a_n \rightarrow +\infty$ when for every real R there is N such that $n \geq N$ implies $a_n > R$. Similarly, $a_n \rightarrow -\infty$ when for every real R there is N such that $n \geq N$ implies $a_n < R$.

Examples include $n^2 \rightarrow +\infty$, $-n \rightarrow -\infty$, and $a^n \rightarrow +\infty$ for $a > 1$. The sequence $((n!)^{1/n})$ also tends to $+\infty$. An unbounded sequence need not tend to $+\infty$; alternating behavior can prevent eventual domination of every bound.

2.6 Ratio and root tests for sequences

Theorem. Ratio test for sequences. For a positive sequence with $a_{n+1}/a_n \rightarrow \ell$, one has $a_n \rightarrow 0$ when $0 \leq \ell < 1$, and $a_n \rightarrow +\infty$ when $\ell > 1$. When $\ell = 1$, no conclusion follows.

Theorem. Root test for sequences. For a positive sequence with $\sqrt[n]{a_n} \rightarrow \ell$, one has $a_n \rightarrow 0$ when $0 \leq \ell < 1$, and $a_n \rightarrow +\infty$ when $\ell > 1$. The case $\ell = 1$ is inconclusive.

2.7 Subsequences

Definition. If (r_n) is a strictly increasing sequence of natural numbers, then (a_{r_n}) is a *subsequence* of (a_n) . The order of the selected terms is preserved.

A convergent sequence and all its subsequences have the same limit. This gives a powerful way to prove divergence: find two subsequences with distinct limits, or find subsequences whose behavior is incompatible with convergence.

Theorem. Monotone subsequence theorem. Every real sequence has a monotone subsequence.

Bolzano-Weierstrass theorem. Every bounded real sequence has a convergent subsequence.

2.8 Cauchy sequences

Definition. A sequence (a_n) is *Cauchy* if for every $\varepsilon > 0$ there exists N such that

$$|a_n - a_m| < \varepsilon \quad (m, n \geq N).$$

Theorem. A real sequence is convergent if and only if it is Cauchy. Thus, Cauchy behavior characterizes convergence without naming the limit in advance.

3 Introduction to Infinite Series

A series is an infinite formal sum. Its meaning is obtained through finite partial sums; this avoids the ambiguity of grouping infinitely many terms.

3.1 Convergence and sum of an infinite series

Definition. For $\sum_{n=1}^{\infty} a_n$, the n th partial sum is

$$s_n = a_1 + a_2 + \cdots + a_n.$$

The series converges when (s_n) converges. If $s_n \rightarrow s$, then s is the sum of the series.

Theorem. Changing only finitely many initial terms does not change convergence or divergence. If two series converge, their termwise sum and scalar multiples converge with the expected sums.

Theorem. Necessary condition. If $\sum a_n$ converges, then $a_n \rightarrow 0$. The converse is false.

Theorem. Cauchy criterion for series. The series $\sum a_n$ converges if and only if for every $\varepsilon > 0$ there is N such that

$$|a_{n+1} + \cdots + a_m| < \varepsilon \quad (m > n \geq N).$$

Theorem. Absolute convergence. If $\sum |a_n|$ converges, then $\sum a_n$ converges. A convergent series for which $\sum |a_n|$ diverges is conditionally convergent.

Theorem. If $\sum a_n$ converges to s , then its tail can be made arbitrarily small: for every $\varepsilon > 0$ there is N such that

$$\left| \sum_{n=N+1}^{\infty} a_n \right| < \varepsilon.$$

Representative example. For

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)},$$

telescoping gives

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1},$$

so the partial sums reduce to $1 - 1/(n+1)$ and the series sums to 1.

3.2 Tests of convergence of a series of positive terms

For a series with nonnegative terms, the partial sums are increasing. Therefore convergence is equivalent to boundedness of the partial sums.

Theorem. Comparison test. If $0 \leq a_n \leq b_n$ eventually and $\sum b_n$ converges, then $\sum a_n$ converges. If $0 \leq b_n \leq a_n$ eventually and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Theorem. Limit comparison test. If $a_n, b_n > 0$ and

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L, \quad 0 < L < \infty,$$

then $\sum a_n$ and $\sum b_n$ either both converge or both diverge.

Theorem. p -series test.

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges for $p > 1$ and diverges for $0 < p \leq 1$.

Theorem. d'Alembert ratio test. For $a_n > 0$, let

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}.$$

If $L < 1$ the series converges; if $L > 1$ the series diverges; if $L = 1$ the test is inconclusive.

Theorem. Cauchy root test. For $a_n \geq 0$, let

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{a_n}.$$

If $L < 1$ the series converges; if $L > 1$ the series diverges; $L = 1$ is inconclusive.

Definition. For functions on $\mathbb{N}$, the notation $f(n) = O(g(n))$ means that there is a constant $C > 0$ such that $|f(n)| \leq C|g(n)|$ for all sufficiently large n .

Theorem. Gauss's test. Let $\sum a_n$ be a positive-term series and suppose

$$\frac{a_n}{a_{n+1}} = 1 + \frac{\alpha}{n} + \frac{b_n}{n^p}, \quad p > 1,$$

where (b_n) is bounded. Then $\sum a_n$ converges when $\alpha > 1$ and diverges when $\alpha \leq 1$. An equivalent asymptotic form is

$$\frac{a_n}{a_{n+1}} = 1 + \frac{\alpha}{n} + O\left(\frac{1}{n^p}\right), \quad p > 1.$$

These tests are complementary: comparison is best for direct inequalities, limit comparison for asymptotically equivalent terms, the ratio/root tests for factorials and exponentials, and Gauss's test for ratios approaching 1 too closely for d'Alembert's criterion.

3.3 Tests of convergence for Alternating Series

Theorem. Leibniz alternating-series test. If $a_n \geq 0$, $a_{n+1} \leq a_n$, and $a_n \rightarrow 0$, then

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n$$

converges.

Theorem. Abel's test. If $\sum a_n$ converges and (b_n) is monotone and bounded, then $\sum a_n b_n$ converges.

The distinction between absolute and conditional convergence is essential. For example, the alternating harmonic series converges by Leibniz's test but is not absolutely convergent because the harmonic series diverges.

4 Limits

The material develops finite limits of functions using an ε - δ definition, one-sided limits, and a sequential characterization, followed by infinite limits.

4.1 Finite Limits

Definition. Let $f : J \setminus \{a\} \rightarrow \mathbb{R}$. We say $\lim_{x \rightarrow a} f(x) = \ell$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x) - \ell| < \varepsilon.$$

Theorem. A finite function limit, when it exists, is unique. Limits obey the algebraic rules for sums, scalar multiples, products, and reciprocals when the limiting denominator is nonzero.

Definition. The right-hand limit $\lim_{x \rightarrow a^+} f(x) = \ell$ requires $0 < x - a < \delta$. The left-hand limit $\lim_{x \rightarrow a^-} f(x) = \ell$ requires $0 < a - x < \delta$.

Theorem. The two-sided limit exists and equals ℓ exactly when the left-hand and right-hand limits both exist and equal ℓ .

Theorem. Sequential characterization. $\lim_{x \rightarrow a} f(x) = \ell$ if and only if, for every sequence (x_n) in $J \setminus \{a\}$ with $x_n \rightarrow a$, one has $f(x_n) \rightarrow \ell$.

Representative example. Since polynomials are built from functions with known limits, $\lim_{x \rightarrow 2} (x^2 + 1) = 5$ follows immediately from the algebra of limits.

4.2 Infinite Limits

Definition. We write $\lim_{x \rightarrow a} f(x) = +\infty$ when, for every $M > 0$, there is $\delta > 0$ such that $0 < |x - a| < \delta$ implies $f(x) > M$. The definition for $-\infty$ is analogous with $M < 0$ and $f(x) < M$.

Representative example. Given $M > 0$, choosing $\delta = 1/\sqrt{M}$ shows that $0 < |x - 2| < \delta$ implies $1/(x - 2)^2 > M$, hence $1/(x - 2)^2 \rightarrow +\infty$ as $x \rightarrow 2$.

5 Continuity

Continuity connects the value of a function at a point with its limiting behavior nearby.

Definition. A function $f : J \rightarrow \mathbb{R}$ is continuous at $a \in J$ when for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

Equivalently, $\lim_{x \rightarrow a} f(x) = f(a)$.

5.1 Sequential Criterion for Continuity

Theorem. A function is continuous at a in the sequential sense precisely when $x_n \rightarrow a$ implies $f(x_n) \rightarrow f(a)$ for every sequence in its domain. The sequential and ε - δ definitions of continuity are equivalent.

Theorem. If f and g are continuous at a , then $f + g$, αf , fg , and f/g (when $g(a) \neq 0$) are continuous at a .

6 Discontinuity

A discontinuity occurs when the local behavior required by continuity fails. The course emphasizes three basic ways failure can occur: the limit does not exist, the limit exists but differs from the function value, or the function is not defined at the point.

6.1 Types of Discontinuities

The important local patterns are:

- removable: a finite limiting value exists but does not agree with the assigned value or the point is missing;
- jump: finite one-sided limits exist but are unequal;
- infinite: the function becomes unbounded near the point.

These categories organize how a graph can fail to match the definition of continuity.

6.2 Properties of Continuous Functions

Theorem. Cantor nested interval theorem. A nested sequence of nonempty closed bounded intervals has nonempty intersection when their lengths shrink appropriately as in the completeness framework used in the course.

Theorem. Bolzano's theorem. If $f : [a, b] \rightarrow \mathbb{R}$ is continuous and $f(a)$ and $f(b)$ have opposite signs, then some $c \in (a, b)$ satisfies $f(c) = 0$.

Theorem. Intermediate Value Theorem. If f is continuous on $[a, b]$ and λ lies between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ with $f(c) = \lambda$.

Theorem. Fixed Point theorem. If $f : [a, b] \rightarrow [a, b]$ is continuous, then there exists $c \in [a, b]$ with $f(c) = c$.

Theorem. Weierstrass and Extreme Value theorems. A continuous function on a closed bounded interval is bounded and attains both its maximum and its minimum there.

Representative example. To establish existence of a root, define a continuous function whose endpoint values have opposite signs; Bolzano then supplies a point where the function vanishes.

6.3 Integral Test for Series

Theorem. Integral test. Let $f : [1, \infty) \rightarrow [0, \infty)$ be continuous and decreasing, with $a_n = f(n)$. Then

$$\sum_{n=1}^{\infty} a_n$$

converges exactly when

$$\int_1^{\infty} f(x) dx$$

converges.

For $f(x) = x^{-p}$ this gives the p -series criterion. More generally it applies to terms such as $1/(n(\log n)^p)$ after choosing the corresponding decreasing continuous function.

7 Differentiation

Continuity describes whether a function has a jump or break. Differentiation measures the instantaneous rate of change.

7.1 Differentiability of Functions

Definition. For $f : (a, b) \rightarrow \mathbb{R}$, the function is differentiable at $c \in (a, b)$ when

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

exists as a finite real number. Equivalently,

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h}.$$

Theorem. If a function is differentiable at c , then it is continuous at c . The converse is false; $|x|$ is continuous at 0 but not differentiable there because its left and right difference quotients are unequal.

Theorem. Algebra of differentiation. If f, g are differentiable, then

$$(f + g)' = f' + g', \quad (\lambda f)' = \lambda f', \quad (fg)' = f'g + fg',$$

and, where $g \neq 0$,

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

Theorem. Chain rule. If f is differentiable at c and g is differentiable at $f(c)$, then

$$(g \circ f)'(c) = g'(f(c))f'(c).$$

Representative example. For $g(y) = \sin y$ and $f(x) = x^2$,

$$(g \circ f)'(x) = \cos(x^2) 2x = 2x \cos(x^2).$$

7.2 Applications

Theorem. Rolle's theorem. If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then some $c \in (a, b)$ satisfies $f'(c) = 0$.

Theorem. Mean Value Theorem (Lagrange). If f is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Geometrically, the tangent slope at c equals the slope of the secant joining the endpoints.

Theorem. Constant Function theorem. If f is continuous on $[a, b]$, differentiable on (a, b) , and $f'(x) = 0$ throughout (a, b) , then f is constant on $[a, b]$.

Theorem. Monotonicity theorem. If $f'(x) \geq 0$ on an interval, then f is increasing there; if $f'(x) \leq 0$, then f is decreasing. Strict derivative inequalities give strict monotonicity under the stated differentiability hypotheses.

7.3 Further consequences

Theorem. Cauchy's Mean Value Theorem. If f and g are continuous on $[a, b]$ and differentiable on (a, b) , then for some $c \in (a, b)$,

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

Theorem. Darboux's theorem. Derivatives have the intermediate value property: even when f' is not continuous, it cannot jump from one value to another without taking all intermediate values.

The course also emphasizes that derivatives need not themselves be continuous, and that a differentiable function can have an oscillating derivative near a point.

7.4 Local Extrema

Definition. A point c is a local maximum of f if there is $\delta > 0$ such that $f(c) \geq f(x)$ whenever x lies in the domain and $|x - c| < \delta$. A local minimum is defined with the inequality reversed. Global extrema compare the point against every point of the domain.

Theorem. Fermat's theorem. If c is an interior point at which f is differentiable and c is a local maximum or local minimum, then

$$f'(c) = 0.$$

Boundary points need not satisfy this condition.

Theorem. First derivative test. If f' changes from positive to negative across a critical point, f has a local maximum there. If f' changes from negative to positive, f has a local minimum. A derivative with the same sign on both sides does not produce an extremum.

Theorem. Higher-order derivative test. Suppose the first nonzero derivative at c after $f'(c) = 0$ is $f^{(n)}(c)$. If n is even, the sign of $f^{(n)}(c)$ determines a local minimum or maximum; if n is odd, c is not a local extremum under the stated hypotheses. In particular, $f''(c) > 0$ gives a local minimum and $f''(c) < 0$ gives a local maximum.

Representative example. For $f(x) = |x|$, the point 0 is a local and global minimum, but $f'(0)$ does not exist. This shows why extrema may occur at nondifferentiable points and why endpoint and critical-point checks must be treated separately.

Coverage and notation map

This companion follows the parent sequence of topics: Basics; Sequences and their convergence, monotonicity, sandwich principle, divergence to $\pm\infty$, ratio/root criteria, subsequences and Cauchy sequences; Infinite Series and their convergence tests; Limits and Infinite Limits; Continuity and its sequential criterion; Discontinuity and properties of continuous functions; the Integral Test; and Differentiation, its standard rules, mean-value consequences, and local extrema. The companion intentionally omits the source document's separate question sets.